

Assaad El Halabi

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PROFESSIONAL SUMMARY

Senior Backend and AI Automation Engineer with 6+ years of experience building scalable backend services, cloud infrastructure, and production automation systems using Python, Node.js, TypeScript, and AWS. Strong experience designing AI-powered workflow automation for healthcare and operations, including LLM-based orchestration, OCR/document processing, RAG assistants, voice-agent integrations. Experienced in translating complex business workflows into reliable, observable, and production-ready automation pipelines.

EXPERIENCE

Smarter Technologies (Thoughtful AI)

07/2025 – 04/2026

Forward Deployed Engineer

- Built AI-powered healthcare automation workflows for prior authorization, claims processing, and revenue-cycle operations using Python, Node.js, APIs, and AWS services.
- Developed RAG-based automation workflows allowing AI systems to retrieve relevant healthcare/business context and execute workflow decisions with stronger grounding.
- Developed OCR and document-processing pipelines for physician-filled forms, patient medical-history documents.
- Designed LLM-powered automation pipelines cutting manual revenue cycle intervention by 70%+ across client deployments.
- Engineered custom backend services for complex healthcare workflows, including API orchestration, and secure data transfer.
- Translated client healthcare operations into parallel and concurrency-safe backend services.
- Led HIPAA-aware end-to-end deployments from code to delivery, handling configuration, troubleshooting, and optimization.

Oreveon

07/2022 – 07/2025

Software Engineer III

- Engineered robust containerized Node.js/Python microservices, achieving 99.9% SLA and reducing time-to-insight by 85%.
- Built backend workflows supporting AI/ML models for segmentation, recognition, and object detection with large image datasets.
- Designed data and model-artifact lifecycle workflows covering encryption, storage, shipment, versioning, and metadata tracking.
- Designed AWS microservices using Lambda, API Gateway, S3, RDS, Docker, and Step Functions.
- Automated 90%+ of manual batch-handling tasks via reusable microservices, cutting operational overhead by over 80 hrs/month.
- Built end-to-end CI/CD pipelines with GitHub Actions and AWS CodePipeline—reduced deployment lead time by 60%.
- Implemented CMMC Level 1 and 2 compliance across backend systems and cloud infrastructure.
- Built a platform that accelerated team operations like batch assignment (20 min → 35 s), payment calculations (1 hr → 39 s).
- Collaborated cross-functionally with product, ML, and ops teams to maintain 99.9% uptime for airport-critical applications.

American University of Beirut (AUB)

01/2021 – 06/2022

Software Engineer

- Contributed to Khaddit Beirut initiative: automated data pre-processing and visualization pipelines in Python.
- Developed custom Django dashboards to explore blast-impact datasets, enabling dynamic filtering and geospatial insights.

EDUCATION

AWS Certified Developer – Associate

05/2025 – 05/2028

Credential ID: 7eb3bfd205b844ea990a13bfc6ffc910

OnRamp Lebanon - Fast-Track to Senior Bootcamp

04/2024 – 08/2024

Intensive bootcamp focused on advanced software engineering practices, system design, and leadership skills.

American University of Beirut (AUB)

09/2018 – 02/2024

MS in Computer Science (Full scholarship – MEPI)

BS in Computer Science (Full scholarship – USAID)

PROJECTS

Portfolio RAG Q&A Assistant

06/2026

- Public AI assistant that answers questions about my background, experience, and projects using RAG.

AI Voice Agent / Inbound Sales Automation

05/2026

- Built an AI voice-agent workflow for inbound carrier calls, MC verification, load retrieval, rate negotiation, and booking.
- Implemented real-time APIs for carrier verification, load lookup, offer evaluation, and post-call ingestion.

- Built post-call AI processing for outcome classification, sentiment analysis, transcript review, and 13+ field extraction.
- Developed a live FastAPI/PostgreSQL dashboard with Docker, Railway, API-key auth, and auto-refreshing KPIs.
- Designed backend policy guardrails to enforce pricing rules outside of prompt-only logic.

CollaBoard

05/2024

- Collaborative Whiteboard supporting real-time user interaction. OnRamp Bootcamp's final Project.

AraDialAug

02/2024

- Master's Thesis: Arabic Conversational models powered by meta-learning and paraphrase-augmentation with little data.

JSColab

04/2023

- Zero-setup, notebook-style environment for JavaScript/React prototyping; supports dynamic npm/CSS imports and markdown.

SKILLS

AI / Automation: RAG, LLM workflows, OCR, document processing, AI workflow orchestration, prompt orchestration, structured data extraction, post-call analysis, sentiment classification, healthcare automation, prior authorization, claims automation;

Backend & APIs: Node.js, AWS Lambda, Django, Docker; **Databases**: MySQL, PostgreSQL, SQL Server, AWS RDS;

Cloud: AWS, Azure; **CI/CD & DevOps**: GitHub Actions, Jenkins, Docker, Kubernetes, Ansible, Docker Compose;

Programming Languages: Python, Java, JavaScript, TypeScript, C++, Bash; **Tools**: Git, Svn, Jira, ClickUp, Confluence, Slack.

Languages: Arabic (Native), English (Proficient), Portuguese (Proficient), Spanish (Intermediate)